

Q: I am a student who is a first-time learner of Python and its packages. Please provide an at-a-glance summary of how pandas has been used in the attached marimo notebooks (Wk02_Marimo_ZScore.py, Wk03_InteractivePlots_Plotly.py, Wk04_DataPreparation.py) to help me understand its usage better.

Let's explore how `pandas` is used across your notebooks to build your data analysis skills. I will ask guiding questions along the way to help you master these concepts.

The `pandas` library is a foundational tool for data manipulation in Python. While your first notebook introduces basic Python exception handling without `pandas`, your other notebooks rely heavily on it to process financial data.

Here is an at-a-glance summary of the methods used:

 Category	 Pandas Operations
Importing & Inspecting	Loading CSV files with <code>pd.read_csv()</code> and previewing rows with <code>df.head()</code> .
Cleaning Data	Replacing missing values using <code>.fillna(0)</code> and finding missing data with <code>.isna()</code> .
Filtering & Sorting	Isolating specific companies (e.g., <code>df['Ticker'] == 'MSFT'</code>) and ordering data chronologically with <code>.sort_values()</code> .
Advanced Calculations	Using <code>.groupby()</code> alongside <code>.shift()</code> to calculate lagged values, and applying vectorized math across entire columns for the Z-Score.

To help you get comfortable with these operations, here are three entry points we can explore together:

-  **The Basics of DataFrames:** How `pandas` reads your raw CSV files and structures them into tables.
-  **Data Cleaning and Filtering:** How to handle missing data and isolate the exact rows you want to analyze.
-  **Vectorization and Grouping:** How `pandas` performs math on entire columns at once and calculates year-over-year metrics.

Which of these three areas would you like to dive into first?

After selecting 1:

Let's start with the foundation: the **DataFrame**.

You can think of a `pandas` DataFrame as a highly programmable Excel spreadsheet. It organizes your raw data into a neat, 2-dimensional table with rows (usually representing individual records) and columns (representing different variables or features).

In your notebooks, you bring data into Python using the `pd.read_csv()` function. This command reads a `.csv` file and instantly converts it into a DataFrame, which is typically assigned to a variable named `df`.

Once the data is loaded, it's often too large to view all at once. Printing thousands of rows would crash or clutter your notebook! Instead, `pandas` provides a handy method to quickly inspect the column names and the first few entries to make sure the data loaded correctly.

Based on your work in `Wk03_InteractivePlots_Plotly.py` and `Wk04_DataPreparation.py`, do you recall the name of the method used to preview just the first 5 rows of a DataFrame? (*Hint: It shares its name with the top part of your body!*)

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After selecting 2:

Let's dive into data cleaning and filtering 🗡️. Real-world data is rarely perfect—it often has missing pieces or includes much more information than we actually need to analyze.

Let's look at a specific scenario from your `Wk03_InteractivePlots_Plotly.py` notebook. When looking at financial data, a company might not report their "Total Debt". In `pandas`, this missing data is usually represented as `NaN` (Not a Number).

To fix this, your notebook uses the `.fillna()` method on that specific column:

```
df['Total_Debt'] = df['Total_Debt'].fillna(0)
```

This code tells `pandas` to find any `NaN` values in the 'Total_Debt' column and replace them with `0`.

Thinking about the financial context of your project, why might it make sense to replace a missing debt value with a `0` rather than just deleting that company's row from our dataset entirely?

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